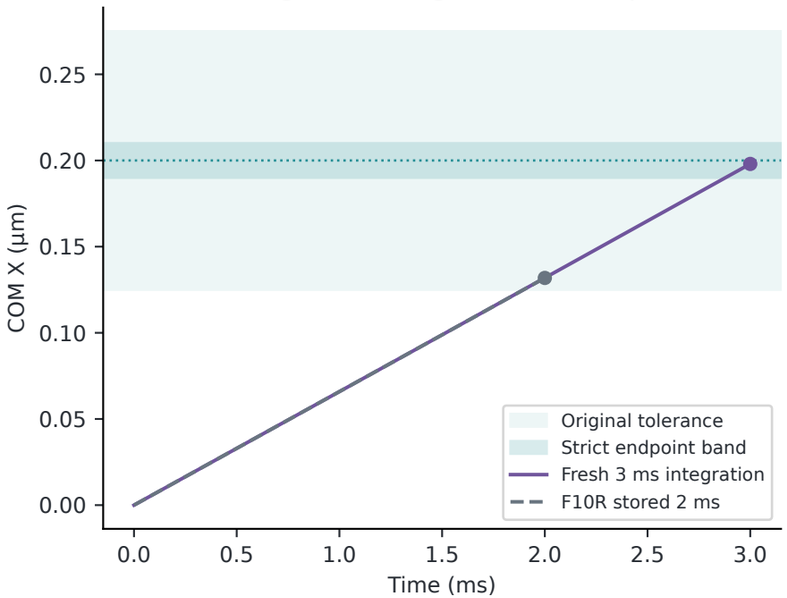
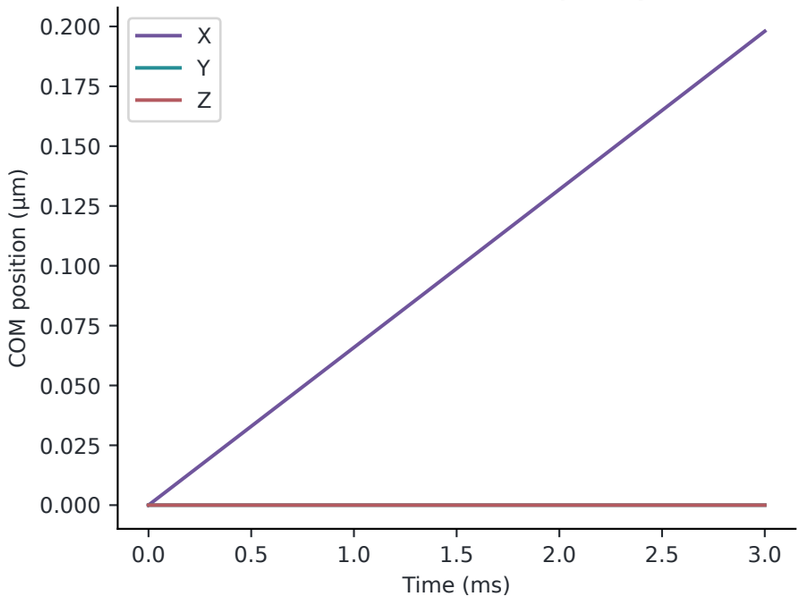
X-pair navigation: fresh 3 ms validation

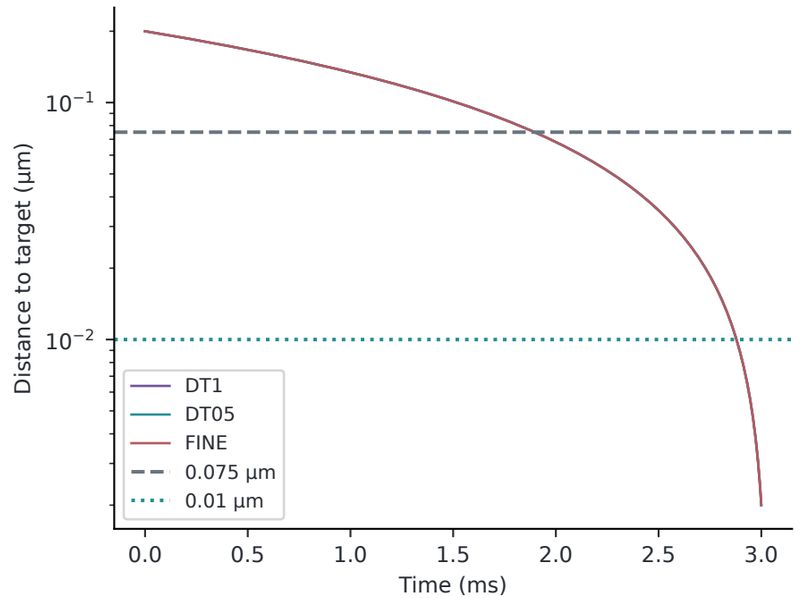
a Registered target: (0.2, 0, 0) µm



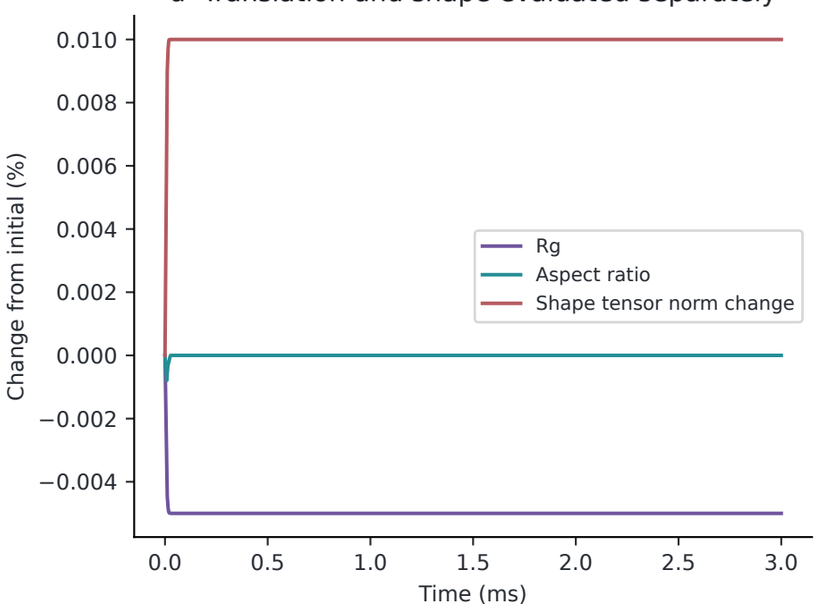
b Actual three-axis trajectory



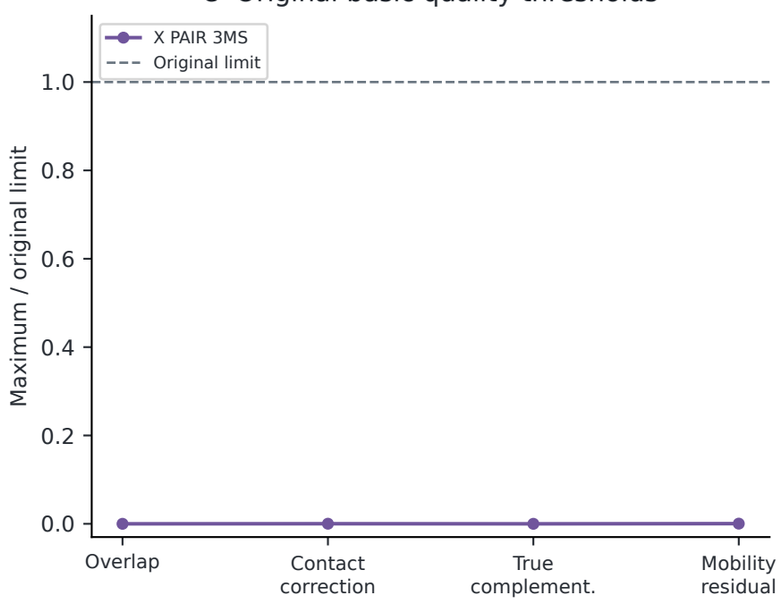
c Endpoint error: 0.002001 µm



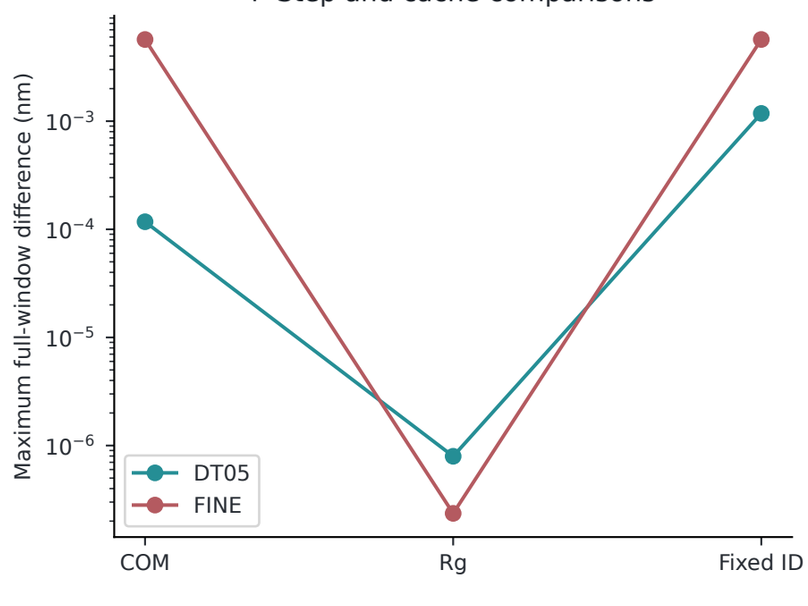
d Translation and shape evaluated separately



e Original basic quality thresholds



f Step and cache comparisons



Same full-precision F10R amplitude/phase; no extrapolated or stitched trajectory · All five numerical layers PASS
EXPLORATORY · HOMOGENEOUS_WATER_CONTROL_PROXY · PHYSICS_UNCALIBRATED · training_ready=false